

KARL HILL

AEROSPACE & MISSION SOFTWARE
SOFTWARE ENGINEERING | TECHNICAL LEADERSHIP | TEAM EXECUTION

SUMMARY

Staff Aerospace Software Engineer and technical delivery leader with experience building mission-critical software and the engineering systems that support its delivery across NASA and national security programs. Lead execution, engineer development, CI/CD and engineering standards for a ~10-engineer team spanning ~20 repositories and multiple deployment environments. Combines hands-on software engineering with team leadership, cross-team coordination, and operational delivery.

PROFESSIONAL EXPERIENCE

Staff Aerospace Software Engineer

SEPT 2025 – PRESENT

Jacobs — National Security · Chantilly, VA

- Lead day-to-day technical delivery for a ~10-engineer team developing Python-based mission software across ~20 repositories and multiple deployment environments; sequence work, coordinate dependencies, and drive integration and release readiness.
- Build and evolve shared engineering systems for CI/CD, automated testing, security gates, repository standards, dependency management, and release automation, improving consistency across independently developed services.
- Develop distributed application integration and messaging capabilities spanning RabbitMQ/ActiveMQ, shared interfaces, service orchestration, and multi-environment deployments.
- Onboarded and coached approximately six engineers through code review, technical feedback, development guidance, and structured growth plans for junior engineers.
- Lead Scrum/Agile execution and coordinate with program stakeholders and external engineering teams to translate mission requirements into executable engineering work and resolve cross-team dependencies.

Lead Software Engineer

DEC 2017 – SEPT 2025

SSAI / NASA Goddard Space Flight Center · Greenbelt, MD

- Architected NASA's cloud-based Flood Mapping System on AWS, delivering near real-time, satellite-derived flood products to support disaster response.
- Modernized NASA LAADS DAAC as Lead Software Engineer — Find Data search and order, the archive portal, and LANCE near-real-time access — and moved delivery onto GitLab CI/CD and Kubernetes.
- Rebuilt NASA Earth Observatory's high-traffic web platform, supporting ~1.5M monthly visitors while improving performance, UX, and SEO.
- Delivered an automated content registry workflow, boosting data collection efficiency by ~60% and accelerating researcher access to new datasets.
- Built a high-performance file and metadata platform on Ceph, improving virtual directory mapping and accelerating discovery for large scientific datasets.

DETAILS

Washington, DC

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LINKS

LinkedIn

linkedin.com/in/khill

GitHub

github.com/karlhillx

Website

karlhill.com

CORE COMPETENCIES

- Software Engineering
- Technical Leadership
- CI/CD & DevSecOps
- Distributed Systems & Messaging
- Engineer Development & Coaching
- Team Execution & Agile Delivery
- Mission Software Delivery

PROFESSIONAL EXPERIENCE

Senior Software Engineer

JAN 2016 — DEC 2017

InformedDNA · Washington, D.C.

- Architected and delivered a Laravel-based case management platform, reducing operational costs by \$30K/year.
- Led CRM enhancements that improved retention and contributed ~15% revenue growth through better lifecycle workflows and reporting.
- Spearheaded platform upgrades and security process improvements, doubling incident response efficiency and strengthening operational readiness.

Senior Software Engineer

JUN 2012 — MAR 2015

Ticomix, Inc. · Washington, D.C.

- Delivered SugarCRM solutions for 20+ clients (including Virginia Department of Transportation, Washington Redskins, and Kastle Systems), improving sales operations and team productivity.
- Drove execution discipline that cut backlog ~90%, improving delivery predictability, product quality, and customer satisfaction.

Earlier Engineering Experience

1997 — 2012

Sabre Corporation · Dante Inc. · Visitar Inc. · Verizon Business

- Held software engineering and principal roles across travel, enterprise CRM, telecommunications, and managed security — delivering production systems for clients including Comcast, Mastercard, Verizon/MCI, and global travel customers.
- Shipped Finium, the multi-tenant managed-security platform that enabled a \$105M acquisition, and helped mature engineering practices around testing, code quality, and cross-functional delivery.

EDUCATION

Bachelor of Science in Computer Science coursework, University of Maryland

Associate of Arts, General Studies — Computer Science Curriculum, Howard Community College

Project Management Certificate, Rutgers University

CERTIFICATIONS

SAFe® Agilist, Scaled Agile (in progress)

Professional Scrum Master™ II, Scrum.org

Professional Scrum Product Owner™ I, Scrum.org

DEVELOPER TOOLING

bb-run — Local Bitbucket Pipelines runner — CI you can execute on a laptop.

pipeguard — Policy-as-code checks for CI/CD standards and deployment safety.

TECHNICAL EXPERTISE

Languages: Python, TypeScript, Java, Bash

Cloud & Kubernetes: AWS, Docker, Kubernetes, Helm/OCI

CI/CD & Developer Tooling: Bitbucket Pipelines, GitLab CI, GitHub Actions, Release Automation

Python Tooling: uv, Ruff, pytest, ty, pre-commit

Integration: REST/OpenAPI, RabbitMQ, ActiveMQ, Distributed Systems

Data: PostgreSQL, Elasticsearch/OpenSearch, Redis